

EXPERIMENT 5(8) - BUILDING AND VALIDATING LOGISTIC MODELS

UCI DIABETES DATASET

	Glucose	BloodPressure	BMI	Outcome
0	182	68	33.38	1
1	131	102	30.95	0
2	172	89	36.37	0
3	94	116	22.95	0
4	186	94	25.57	1

UCI Dataset Shape:
(100, 4)

PIMA INDIANS DIABETES DATASET

	Glucose	BloodPressure	BMI	Outcome
0	88	65	43.18	0
1	104	115	36.85	0
2	147	58	18.29	1
3	183	103	31.06	1
4	167	107	21.22	0

Pima Dataset Shape:
(100, 4)

FEATURES AND TARGET

Independent Variables:

- Glucose
- BloodPressure
- BMI

Target Variable:

- Outcome

TRAIN / TEST SPLIT

UCI Training samples : 80
UCI Testing samples : 20

Pima Training samples : 80
Pima Testing samples : 20

MODEL TRAINING

UCI Logistic Regression Model trained successfully.
Pima Logistic Regression Model trained successfully.

UCI DIABETES DATASET - LOGISTIC REGRESSION RESULTS

Accuracy : 0.6500
Precision : 0.6364
Recall : 0.7888
F1 Score : 0.6667

PIMA INDIANS DIABETES DATASET - LOGISTIC REGRESSION RESULTS

Accuracy : 0.7500
Precision : 0.7500
Recall : 0.8182
F1 Score : 0.7826

UCI CONFUSION MATRIX

[[6 4]
[3 7]]

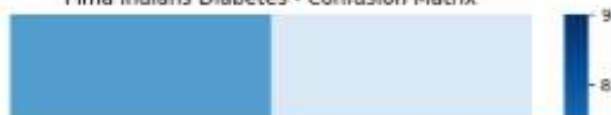
PIMA CONFUSION MATRIX

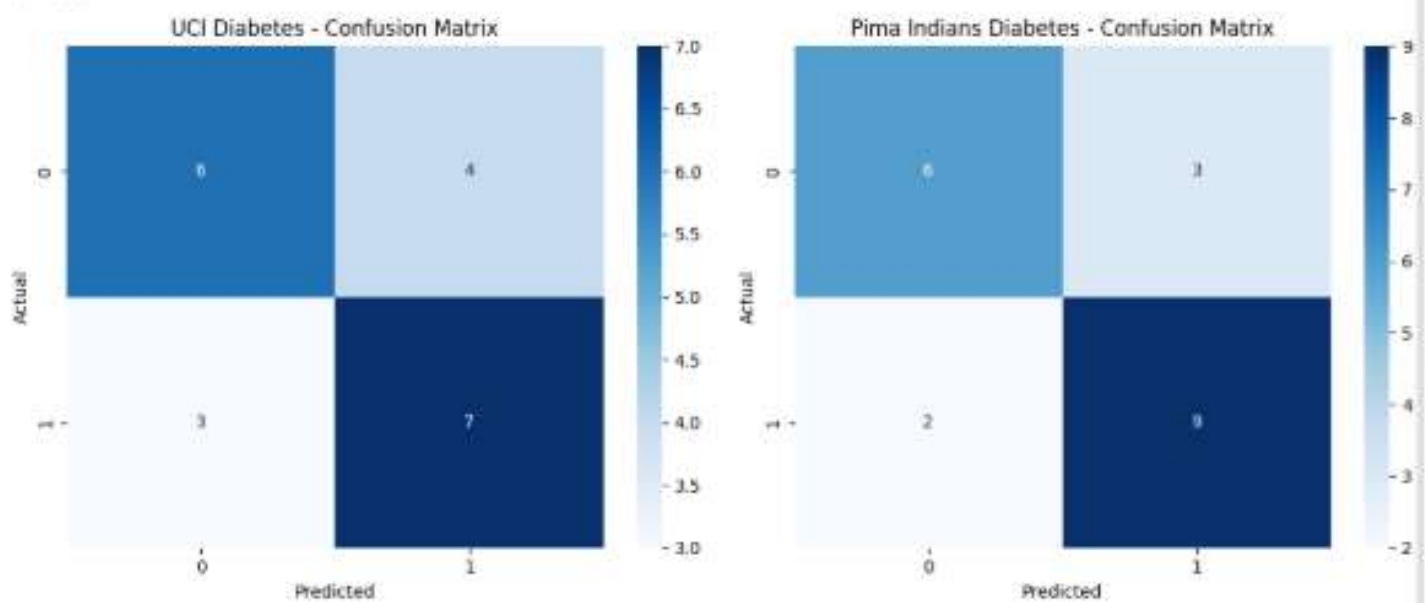
[[6 3]
[2 9]]

UCI Diabetes - Confusion Matrix



Pima Indians Diabetes - Confusion Matrix





FINAL MODEL COMPARISON

Dataset	Accuracy	Precision	Recall	F1 Score
UCI Diabetes	0.65	0.6364	0.7000	0.6667
Pima Indians Diabetes	0.75	0.7500	0.8182	0.7826

RESULT

Logistic Regression models were successfully built and validated for both datasets. Glucose, BloodPressure, and BMI were used to predict diabetes presence. The models were evaluated using Accuracy, Precision, Recall, and F1 Score. Confusion matrices were generated to visualize classification performance.

EXPERIMENT 5(B) COMPLETED SUCCESSFULLY